WHERE BITS MATTER IN WORLD-MODEL PLANNING: A PAIRED MIXED-BIT STUDY OF DINO-WM

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ABSTRACT

Efficient spatial reasoning requires world models that retain planning quality under aggressive quantization. We test whether low-bit behavior is governed mainly by total bitwidth or by *bit allocation* across modules. On DINO-WM planning in Wall, we run paired-goal mixed-bit evaluation with 18 variants, two planner budgets, 65 runs, and 650 episodes. The results show a consistent three-regime structure: 8/6-bit variants remain near FP16, 3-bit variants collapse, and 4-bit is a transition regime where allocation matters. At 4-bit, mixed quantization (encoder FP16, predictor INT4) improves success over uniform INT4 by +0.20 at budget bA (95% bootstrap CI [0.00, 0.40], $n = 30$) and +0.30 at budget bB ([0.00, 0.55], $n = 20$). Asymmetric and layerwise ablations support the same direction: preserving encoder precision is more beneficial than adding predictor precision near the low-bit frontier.

1 INTRODUCTION

World-model planning has become a strong paradigm for sample-efficient control and spatial reasoning, especially when latent dynamics are built on pretrained visual features (Hafner et al., 2023; Oquab et al., 2023; Zhou et al., 2024). Deployment constraints, however, force these models into tight memory and latency budgets where aggressive quantization can destabilize behavior. Recent broad surveys and world-model-specific studies report that low-bit degradation is often non-uniform across modules (Liu et al., 2025; Fu et al., 2026).

This paper asks a concrete workshop question: near low-bit operating points, is planning quality determined mainly by average precision, or by *bit allocation* across encoder and predictor? We study this in DINO-WM on Wall with paired-goal evaluation and mixed-bit variants. Our contribution is empirical but decision-relevant: we identify where low-bit compression remains safe, where it fails, and which allocation choices are most promising. Concretely, we provide three takeaways: (1) a paired mixed-bit frontier showing a repeatable 8/6-safe, 4-transition, 3-collapse pattern; (2) robustness checks across two planner budgets with consistent mixed-vs-uniform INT4 ordering; and (3) encoder-focused evidence from asymmetric and layerwise ablations that sharpens future hypotheses.

2 SETUP

Task and metric. We use pretrained DINO-WM (Zhou et al., 2024) on Wall and keep the repository-native planning success metric from planning logs. We do not redefine task success. All variants use weight-only linear-layer quantization and identical checkpoints.

Variants. We evaluate FP16, uniform INT$\{8, 6, 4, 3\}$, mixed INT$\{8, 6, 4, 3\}$ (encoder FP16, predictor quantized), asymmetric variants (E8/P4, E6/P4, E4/P8, E4/P6), and an INT4 layerwise encoder-retention sweep (0%, 25%, 50%, 75%, 100% encoder kept FP16; predictor fixed INT4).

Evaluation protocol. We use paired-goal evaluation: goals are generated once per seed and reused across variants. Budget bA uses $(\texttt{goal\_H} = 9, \texttt{opt\_steps} = 2, \texttt{max\_iter} = 2)$ with 3 seeds. Budget bB uses $(12, 3, 3)$ with 2 seeds. Each run has 10 episodes, yielding 65 runs and 650 episodes

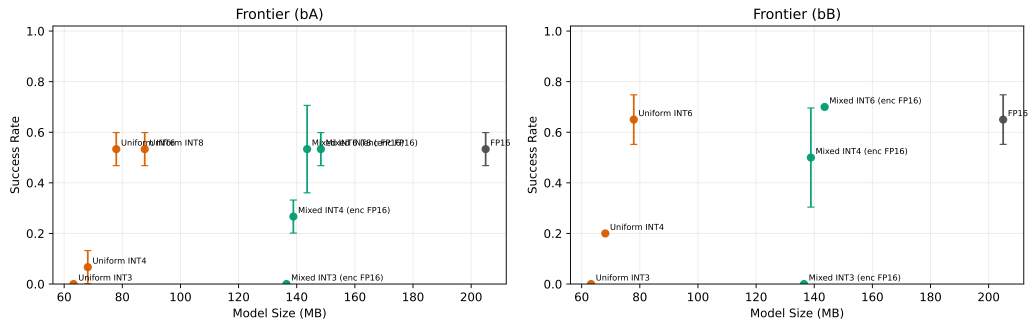

Figure 1: Success–size frontier for the paired mixed-bit study (budgets bA and bB). The 8/6-bit region is stable, the 3-bit region collapses, and the 4-bit region is allocation-sensitive (mixed above uniform). Error bars are run-level 95% confidence intervals.

Variant	Success bA	Success bB	Size (MB)
FP16	0.533	0.650	204.99
Uniform INT6	0.533	0.650	77.92
Mixed INT6	0.533	0.700	143.58
Uniform INT4	0.067	0.200	68.12
Mixed INT4	**0.267**	**0.500**	138.84
Uniform INT3	0.000	0.000	63.23
Mixed INT3	0.000	0.000	136.47

Table 1: Core mixed-bit outcomes from `my_mixedbit_run`. The central effect is the mixed-vs-uniform gap at 4 bits under both planner budgets.

total. We report success, model size, runtime, paired deltas (bootstrap CI), and log-derived mechanistic signals.

3 Main Results

A three-regime pattern appears in one run. Figure 1 and Table 1 show a stable regime at 8/6 bits, a transition at 4 bits, and collapse at 3 bits. At budget bA, FP16, uniform INT8, mixed INT8, and uniform INT6 all reach 0.533 success. At budget bB, FP16 and uniform INT6 both reach 0.650. In contrast, all 3-bit variants are 0.0 at both budgets.

4-bit is allocation-sensitive. At 4 bits, mixed INT4 clearly outperforms uniform INT4: 0.267 vs 0.067 at bA, and 0.500 vs 0.200 at bB. Paired analysis gives +0.20 at bA (95% CI [0.00, 0.40]) and +0.30 at bB ([0.00, 0.55]). Sign-test p-values are 0.109 in both budgets, indicating consistent directional wins with limited non-tie sample size.

Asymmetric and layerwise evidence supports encoder sensitivity. At bA, E6/P4 reaches 0.300 success at 73.19 MB, improving over uniform INT4 (0.067 at 68.12 MB), with paired delta +0.233 (95% CI [0.033, 0.433]). By contrast, increasing predictor bits while keeping encoder at INT4 (E4/P8, E4/P6) does not match mixed INT4. Layerwise INT4 ablation (Appendix, Figure 3) is non-monotonic but peaks when the encoder is fully preserved (0.25 at 100% FP16), reinforcing that encoder precision is the dominant bottleneck near the 4-bit frontier.

Mechanistic diagnostics are consistent with representation degradation. Across variants, success correlates negatively with divergence metrics from planning logs: Spearman $\rho = -0.928$ for mean state distance and $\rho = -0.708$ for visual-embedding divergence (Appendix Figure 5). This does not prove causality, but it supports a plausible mechanism: low-bit encoder degradation weakens latent geometry used for goal-directed planning.

4 DISCUSSION, LIMITATIONS, AND OUTLOOK

The data supports a narrower claim than "mixed is always better": near low-bit transition points, planning quality depends strongly on where precision is allocated. The asymmetric results raise a practical design question for future work: can budget-constrained bit-allocation be optimized directly for planning success rather than model reconstruction error? The non-monotonic layerwise curve suggests a second question: which encoder blocks are precision-critical, and does that set change across environments? Finally, the strong divergence-success correlations motivate causal follow-ups, such as representation-preserving calibration or geometry-aware quantization objectives.

This study is limited to one environment and one checkpoint family. Confidence intervals on key paired effects still include zero because the paired sample is modest. We evaluate weight-only quantization without activation quantization, retraining, or hardware-fused low-bit kernels. Therefore, conclusions should be read as robust empirical hypotheses for efficient spatial reasoning, not universal laws.

5 CONCLUSION

In a paired mixed-bit study on DINO-WM planning, we find that low-bit behavior is structured: 8/6 bits are mostly safe, 3 bits collapse, and 4 bits require careful bit allocation. The most consistent effect is mixed INT4 outperforming uniform INT4 across two planner budgets. These findings make a concrete case for module-aware quantization policies in world-model planning and define targeted follow-up questions for the ES-Reasoning agenda.

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A Appendix: Extended Results with Context

A.1 How to Read the Appendix

The main paper establishes the core claim from one frontier figure and one compact table. This appendix provides supporting analyses that answer three reviewer-facing questions: **(i)** Is the 4-bit effect robust to planner budget? **(ii)** Does evidence localize precision sensitivity to the encoder? **(iii)** Are observed gains consistent with a representation-degradation mechanism?

A.2 Paired Mixed-Bit Effect Table

Table 2 reports paired deltas for the most relevant comparisons. The first two rows are the primary claim (mixed INT4 vs uniform INT4 at bA and bB). The asymmetric rows test whether increasing encoder bits or predictor bits is more useful at similar model sizes.

Comparison	n pairs	Delta	95% CI	Sign-test p
bA: mixed_int4 - uniform_int4	30	+0.200	[0.000, 0.400]	0.109
bB: mixed_int4 - uniform_int4	20	+0.300	[0.000, 0.550]	0.109
bA: enc6_pred4 - uniform_int4	30	+0.233	[0.033, 0.433]	0.065
bA: enc4_pred8 - mixed_int4	30	-0.133	[-0.333, 0.067]	0.344

Table 2: Paired effect estimates from the consolidated mixed-bit pairwise statistics file.

A.3 Budget Robustness Context

Figure 2 shows that the mixed-vs-uniform INT4 gap persists when planner budget increases from bA to bB. This reduces the likelihood that the observed gap is only an artifact of one optimization budget.

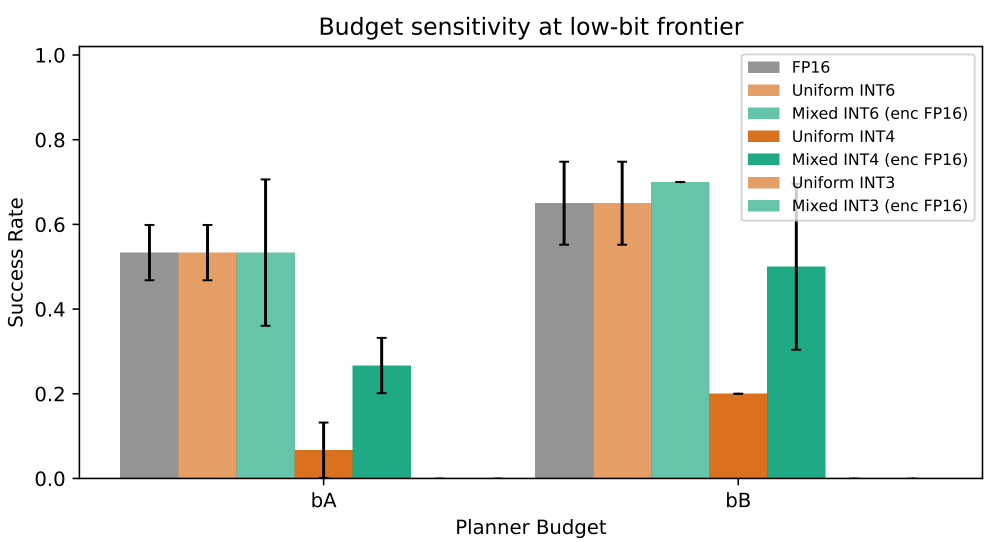

Figure 2: Budget robustness view. Uniform INT3 remains collapsed; mixed INT4 remains above uniform INT4 at both budgets.

A.4 Encoder-Localization Context

Figure 3 gives a coarse localization test: with predictor fixed at INT4, preserving more encoder precision improves peak success, although intermediate points are non-monotonic. This supports the main claim while motivating finer block-level studies.

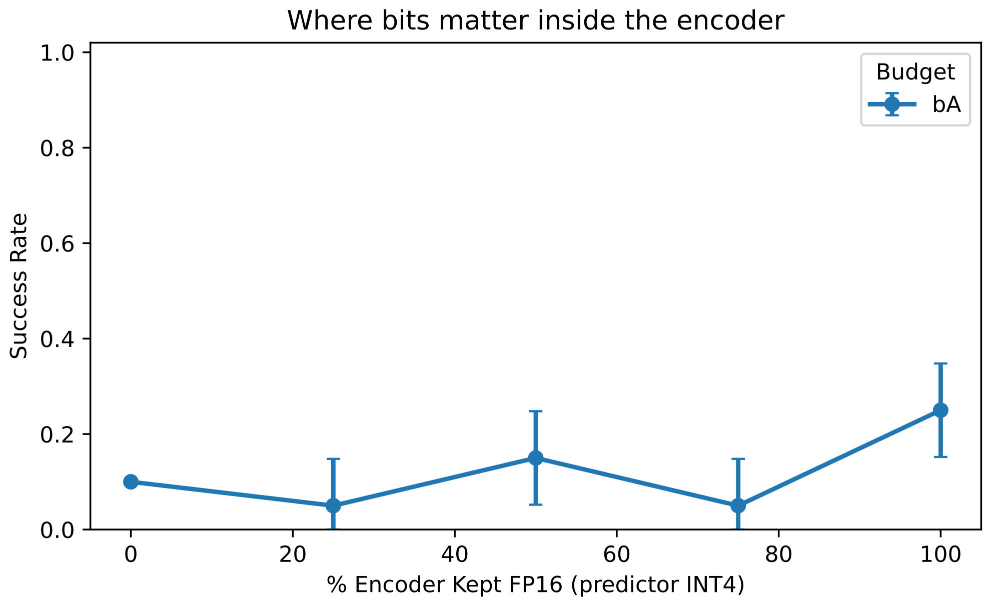

Figure 3: Encoder-retention sweep (INT4 predictor). The highest mean success occurs when encoder precision is fully preserved.

A.5 Difficulty-Conditioned Context

Figure 4 bins episodes by initial goal distance. Mixed INT4 improves over uniform INT4 across bins, suggesting the 4-bit allocation effect is not confined to a single difficulty slice.

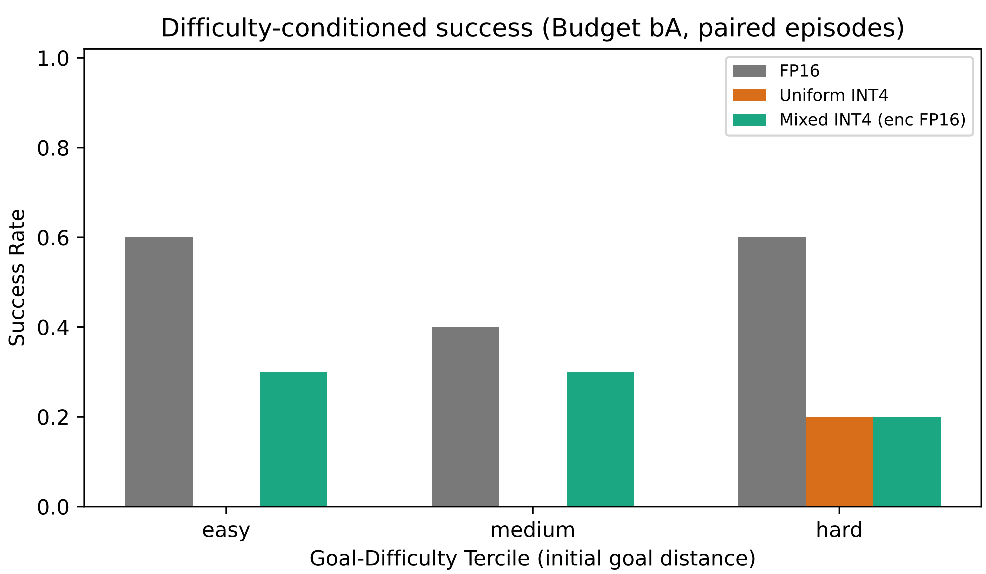

Figure 4: Difficulty-conditioned success at bA (paired episodes). Mixed INT4 consistently improves over uniform INT4.

A.6 Mechanistic Context

Figure 5 links variant-level success to divergence metrics extracted from planning logs. The negative trend supports a representation-degradation explanation for low-bit failure near the 4-bit boundary.

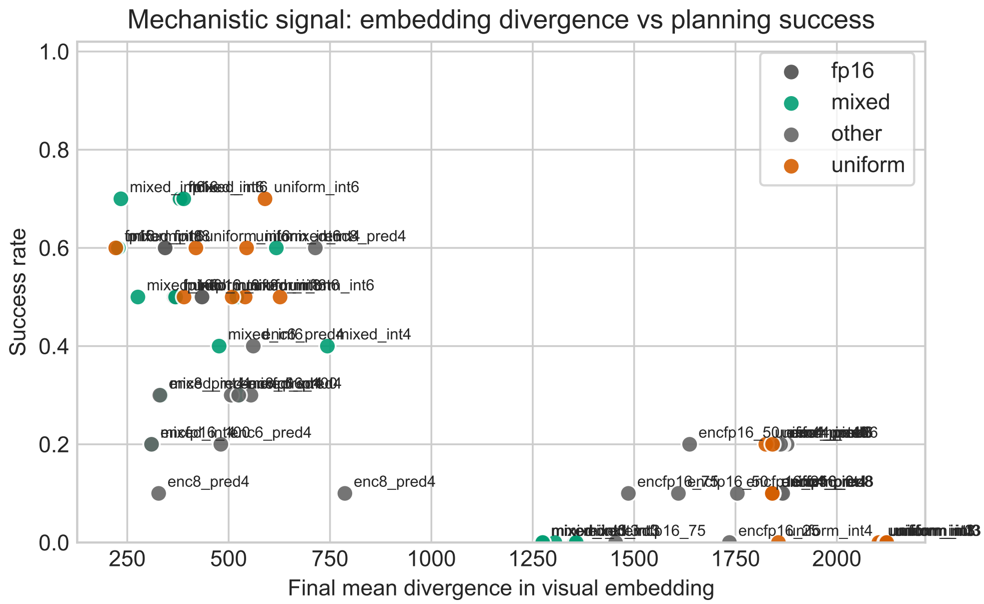

Figure 5: Mechanistic scatter: larger visual-embedding divergence is associated with lower planning success.

A.7 Mixed-Bit Maps for Hypothesis Generation

The next two figures are exploratory and hypothesis-generating. Figure 6 visualizes the uniform/mixed bit ladder directly; Figure 7 maps asymmetric encoder/predictor allocations. Together they motivate a budgeted bit-allocation optimization problem for future work.

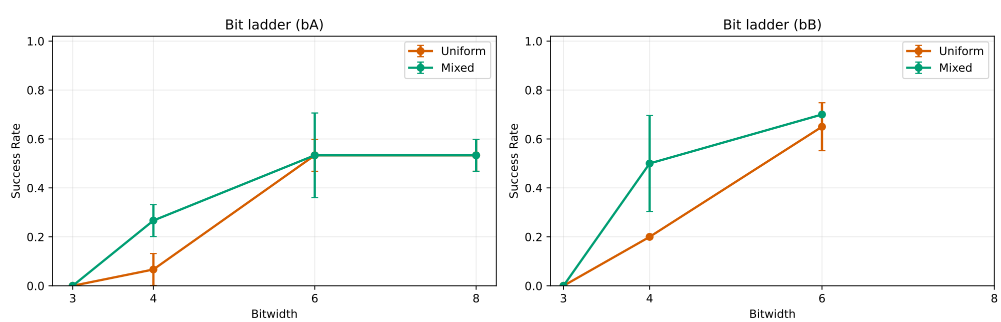

Figure 6: Uniform-vs-mixed bit ladder across budgets. The 4-bit region is where allocation effects are largest.

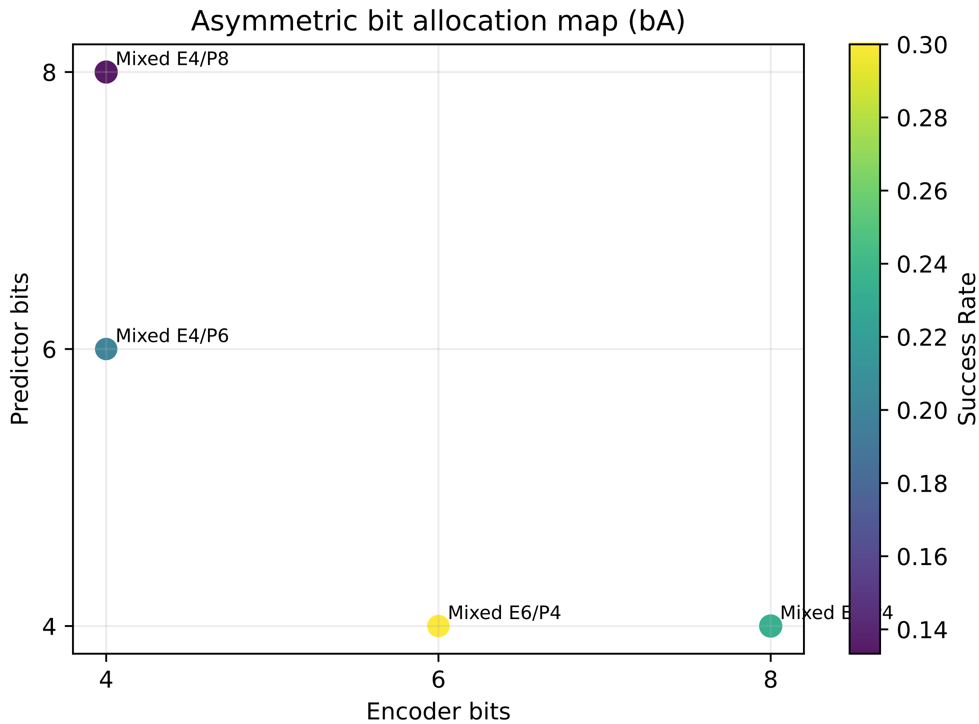

Figure 7: Asymmetric allocation map at bA. Higher encoder precision at fixed predictor precision is generally more beneficial than the reverse.

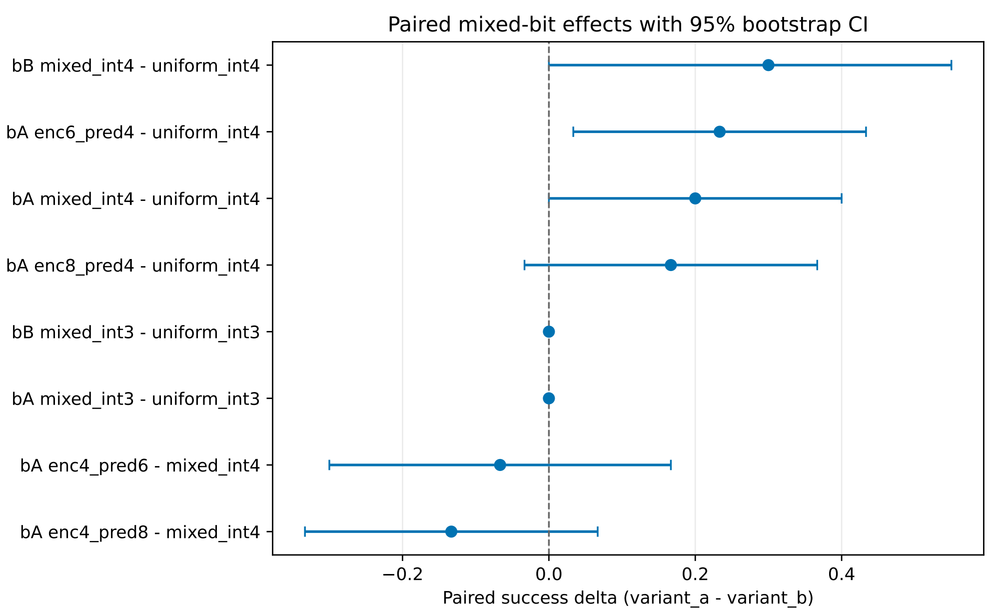

Figure 8: Forest plot of paired deltas with bootstrap confidence intervals for key comparisons.